The Shadow Price of Neutrality

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***Abstract*—TODO(Isaiah), Abstract, 0.2 page (claim → evidence):**

- **Problem: a long/short pod under a platform's neutrality mandate at a fixed 3%/yr risk budget → Sec. II.**
- **Approach: duals of the convex program at a root-found λ^*; three integer sensitivity estimators against certified secant intervals; big-M against perspective formulations → Sec. III.**
- **Q1 finding: the liquidity cap, sector bounds and trading costs carry the cost; factor neutrality is nearly free for α_{mom}; at zero, sector and all-style neutrality cost about the same → 0.16, 0.014, 0.16, 0.961 against 0.956, 0.964; Figs. 1–2.**
- **Q2 finding: the fixed-integer dual is exact when neither the support nor the active set changes within the step (max error 6.7×10^{-12}) and lies in $[L, H]$ in 4 of 23 binding cases; the best estimator depends on the step: at 2017-12 ($n = 11$) the fixed-integer dual is best at $\delta = 5\%$ (0.014; perspective 0.052, big-M 0.24) and the perspective slope at 10% (0.016 against 0.023; big-M 0.073) → Table III.**
- **Q3 finding: two mechanisms. The default-scaling root bound is Gurobi's outer approximation stopping early (29 of 36; 0.015 against 5.7×10^{-4} with a QCP node relaxation); the cone tolerance inflates incumbents (6.4×10^{-3}). Rescaling the weights removes both; presolve, not the formulation, closes big-M's gap → 3.3×10^{-4}, 0.015, 6.0×10^{-4} against 0.057; Fig. 4.**

I. Introduction

TODO(Isaiah), I, 0.6 page (claim → evidence):

- Motivation and significance: a multi-manager platform imposes neutrality to protect its other books; each constraint costs the pod expected alpha; the dual is the price of that constraint → Sec. II.
- Related work: fundamental law and transfer coefficient [1], [2]; constraint attribution [3]; factor alignment [4], [5]; realistic shorts [6]; perspective reformulations [7], [8]; integer sensitivity [9], [10]; sparse portfolios [11]; course texts [12], [13].
- Questions Q1–Q3 → Sec. II-G.
- Contribution: costs at equal risk via duality at λ^*, with the alignment closed form $\sqrt{1-\varphi}$ → A1–A4, Fig. 2.
- Contribution: certified secant intervals that test integer sensitivity estimators against solver error → A8, Table III, 3.6×10^{-4}.
- Contribution: a support-stability account of when the fixed-integer dual is exact → 6.7×10^{-12} (P1), 0.056 (P3).
- Contribution: the perspective model's two scaling mechanisms kept apart, and weight rescaling as the fix → A9, Sec. IV-C, 29 of 36, 1.9×10^{-11}.
- Project type: problem formulation, reusing published formulations in a new setting.

II. Problem Setting

A. Decision problem

A *pod*, one long/short equity team at a multi-manager platform, receives each month a forecast α_i of every stock's *alpha*, its expected return beyond what its factor exposures explain, and chooses weights w as fractions of assets under management (AUM, \$1B). The platform caps the pod's ex-ante volatility at 3%/yr and imposes a mandate that protects its other books (the other pods' portfolios). A portfolio's *factor exposure* is $y = B^\top w$, where B holds each stock's loadings on six factors; its *beta* is y_{mkt}, the exposure to the market factor. *Dollar neutrality* bounds the net weight $|1^\top w|$, *beta neutrality* $|y_{\text{mkt}}|$, style neutrality the other five exposures y_k, and *sector neutrality* each sector's net weight $|S_j^\top w|$, where S_j is the 0/1 indicator of the stocks in GICS sector j (the standard industry classification). *Gross leverage* $1^\top(w^+ + w^-)$ is the long plus the short weight. A stock's *ADV* is its median daily dollar volume over the last 63 trading days, and the *liquidity cap* limits each position to the fraction θ of ADV: $u_i = \min(u, \theta\,\text{ADV}_i/\text{AUM})$. Shorts pay a *borrow cost* c per month and every trade a one-way *trading cost* κ (1 bp $= 0.01\%$); the pod starts from cash ($w_0 = 0$), so the trading cost is κ times gross. Table I lists the base mandate.

B. Data and risk model

Daily adjusted prices and volumes of the current S&P 500 constituents (500 tickers, 2005-01-04 to 2025-12-31) come from yfinance, sectors from the constituent list, and the daily Fama–French five factors and momentum from the French data library [14]–[16]. A day is valid for a stock if it has a return and a dollar volume of at least \$1M; a stock is *eligible* at a date if at least 80% of the 252 trading days ending there are valid. At each of the 241 month-ends from 2005-12 to 2025-12, each eligible stock's daily excess returns over that window are regressed on the six factors. B holds the loadings, $D = \text{diag}(d)$ the residual variances and F the sample factor covariance, both daily values multiplied by 21 trading days to monthly, with d floored at its 5th cross-sectional percentile, so $\Sigma = BFB^\top + D \succ 0$. The eligible universe has 373–499 names.

C. Alphas and alignment

The momentum score is each stock's return over the 252 trading days before the date, skipping the last 21, so it exists at 240 of the 241 month-ends (not the first). It is z-scored across the eligible stocks, winsorized at ± 3 and re-standardized, and turned into an alpha by the forecasting rule $\alpha_i = \text{IC} \cdot \omega_i z_i$ with information coefficient $\text{IC} = 0.03$ and residual volatility

$\omega_i = \sqrt{d_i}$ [1]. It is an input signal; no backtest is run. A D^{-1}-weighted regression of α on the loadings splits it into a residual $\alpha_\perp = \alpha - B(B^\top D^{-1} B)^{-1} B^\top D^{-1}\alpha$ and a factor-spanned part $\alpha_\parallel = \alpha - \alpha_\perp$, both rescaled to α_{mom}'s *information ratio* $\sqrt{\alpha^\top \Sigma^{-1}\alpha}$ (the best ratio of expected alpha to ex-ante risk). The *alignment share* $\varphi = \alpha_\parallel^\top \Sigma^{-1}\alpha_\parallel / \alpha^\top\Sigma^{-1}\alpha \in [0,1]$ is the factors' part of it. The *transfer coefficient* $\mathrm{TC} = \alpha^\top w / (\sqrt{\alpha^\top\Sigma^{-1}\alpha}\sqrt{w^\top \Sigma w})$ is the Σ-metric form of Clarke et al.'s transfer coefficient [2].

D. The convex problem P-RB

With $w = w^+ - w^-$, $w^\pm \geq 0$, and σ_m the monthly volatility budget,

$$\begin{aligned} \max\ & \alpha^\top w - c\,1^\top w^- - \kappa\,1^\top(w^+ + w^-) \\ \text{s.t. }\ & y^\top F y + \textstyle\sum_i d_i w_i^2 \leq \sigma_m^2, \quad y = B^\top w, \\ & |1^\top w| \leq \varepsilon_{\mathrm{net}},\ |y_{\mathrm{mkt}}| \leq \varepsilon_\beta,\ |y_k| \leq b_k,\ |S_j^\top w| \leq s_j, \\ & 1^\top(w^+ + w^-) \leq G, \quad w_i^+, w_i^- \leq u_i. \end{aligned} \tag{1}$$

Because $c, \kappa > 0$, at most one of $w_i^\pm$ is positive at an optimum. The optimal value A^* is net expected alpha per month. Each row k has a multiplier μ_k, and a mandate parameter p (a bound, u or θ) has the *family dual* $\mu_p = dA^*/dp = \sum_k \mu_k\, \partial r_k/\partial p$ over the rows whose right-hand side r_k moves with p (A5): θ through every θ-limited cap ($\partial u_i/\partial\theta = \mathrm{ADV}_i/\mathrm{AUM}$), u through every cap equal to u, a common sector or all-styles bound through all its rows; each sector and style factor is also a key of its own. The *elasticity* $e_p = \mu_p p / A^*$ is the percent change in alpha per percent change in p, and a constraint *binds* if $e_p \geq 10^{-4}$. The risk row's multiplier is λ^*: the price of risk at which a mean-variance investor picks the same portfolio (A1). Fixing the risk budget rather than λ compares costs at equal risk, so relaxing a constraint cannot buy alpha by levering up.

E. The integer problem P-MI

P-MI adds a cardinality limit K and a minimum position $\ell = 0.25\%$, moves the risk row into the objective at λ^*, and raises the position cap to $u = 6\%$:

$$\begin{aligned} \max\ & \alpha^\top w - \lambda^*\big(y^\top F y + \textstyle\sum_i d_i w_i^2\big) \\ & \quad - c\,1^\top w^- - \kappa\,1^\top(w^+ + w^-) \\ \text{s.t. }\ & \text{the mandate of (1) without the risk row,} \\ & \ell z_i^\pm \leq w_i^\pm \leq u_i z_i^\pm,\ z_i^+ + z_i^- \leq 1,\ \textstyle\sum_i (z_i^+ + z_i^-) \leq K, \\ & z^\pm \in \{0,1\}^N. \end{aligned} \tag{2}$$

λ^* is P-RB's multiplier on the same universe at the base mandate with $u = 6\%$, held fixed for that date and universe: a Lagrangian relaxation of the risk row, with zero gap in the convex case. The *big-M* formulation keeps $\sum_i d_i w_i^2$ with $M = u_i$. The *perspective* formulation replaces it by $\sum_i d_i s_i$ with rotated cones $w_i^2 \leq s_i\zeta_i$, $s_i \geq 0$, $\zeta_i = z_i^+ + z_i^-$. The *rescaled perspective* divides every weight, row and bound by $\max_i u_i$ and writes each cone in its own name's bound, $t_i = w_i/u_i$, $s_i' = s_i/u_i^2$, $t_i^2 \leq s_i'\zeta_i$; its feasible set and objective are unchanged, only the quantities Gurobi's absolute tolerances act on differ (A9). With λ^* fixed, P-MI's ex-ante volatility drifts from the budget (computed from the stored solutions as part of N3): median 2.87%/yr in I1 (2.42–3.02, $n = 144$), 2.91% in I2 (2.85–2.96, $n = 82$) and 2.77% in N1 (2.22–3.31, $n = 36$ in each arm, 4 above). Q2's slopes are thus slopes of the risk-penalized objective, equal to equal-risk slopes only at the convex base point, where the baseline (P-MI with $\ell = 0$ and no cardinality row) is P-RB (envelope theorem; validated below).

TABLE I

BASE MANDATE (P-RB) AND THE P-MI PARAMETERS. STYLE BOUNDS ARE ON FACTOR EXPOSURES (LOADING UNITS); MOMENTUM IS ONE OF THE FIVE STYLES.

Parameter	Base
Risk budget (ex-ante volatility)	3%/yr
Gross leverage G	2
Net (dollar) $\varepsilon_{\mathrm{net}}$	$\pm$2%
Beta ε_β	$\pm$0.05
Each style b_k	$\pm$0.10
Each sector s_j (net)	$\pm$2%
Position cap u (P-MI)	3% (6%)
Liquidity θ	5% of ADV
AUM	\$1B
Borrow c	30 bp/yr
Trading κ (one-way)	5 bp
Minimum position ℓ (P-MI)	0.25%

F. Problem classes

P-RB is a convex quadratically constrained program (QCQP); its risk row is the second-order cone $\|(L_F^\top y, \sqrt{d}\circ w)\|_2 \leq \sigma_m$ with $F = L_F L_F^\top$. When $\lambda^* > 0$ its optimal w is unique: by A1, w also maximizes the mean-variance objective at $\lambda = \lambda^*$ over the mandate, which is strictly concave in w ($\Sigma \succ 0$, and the costs are convex and piecewise linear in w); the split $w^\pm$ is then unique because $c, \kappa > 0$. The big-M P-MI is a convex mixed-integer QP (every node relaxation a convex QP); the perspective P-MI is a mixed-integer second-order cone program.

G. Questions

Q1 (convex): at a fixed risk budget, which mandate constraints bind, what does each cost, and what determines that cost? **Q2** (integer): once a cardinality limit and a minimum position enter, how well do continuous sensitivity tools (relaxation duals, fixed-integer duals) predict the true marginal cost of a constraint? **Q3** (formulation and numerics): how do the big-M and perspective formulations compare as the mandate tightens, and how do solver tolerances interact with problem scaling?

III. Solution Approach and Analysis

A. Experiments

Table II defines the experiments: C1–C2 answer Q1, I1–I2 Q2, N1–N3 Q3. D12 is 12 month-ends (2008-10, 2009-03, 2011-08, 2013-06, 2015-08, 2017-12, 2018-12, 2020-03, 2020-11, 2022-06, 2023-03, 2025-04) spanning calm and stressed markets. C1's 72 idealized solves are two mandates

TABLE II

EXPERIMENTS. SOLVES: ROWS OF EACH RUN'S MAIN SOLVE TABLE (C1: BASE AND IDEALIZED; I1, I2, N1, N2(A): MIQPS; N2(B): QCQPS; N3: ROOT-ONLY PROBES). ENVELOPE STENCILS, BASELINE SOLVES AND RELAXATIONS ARE NOT COUNTED.

exp.	instances	varied, arms	solves
C1	D12 (12 dates); all eligible names; α_{mom}, $\alpha_\perp$, $\alpha_\parallel$	base mandate, and the idealized mandates (i), (ii); envelope at $\delta = 1\%$ (0.1%)	108
C2	D12; all eligible; α_{mom} ($\alpha_\perp$, $\alpha_\parallel$ on factor families)	one family swept over its grid, the others at base	2040
I1	2009-03, 2017-12, 2020-03; $N = 250$; $f \in \{0.25, 0.5, 0.75\}$	4 bounds at $\delta = 10\%$ (5% at 2017-12); big-M and rescaled perspective	288
I2	2017-12 ($K = 67$); 2020-03 ($K = 79$); $N = 250$	sector bound at 41 points on $[0, 2b]$; as I1	164
N1	2017-12, 2020-03; $N \in \{100, 250, all\}$; $f \in \{0.25, 0.5\}$; 3 tightness levels	big-M, perspective (default scaling, rescaled); presolve off at $N = 250$; 3 reruns	147
N2(a)	N1 at $N = 250$, base tightness	the N1 arms at $s_{\mathrm{obj}} \in \{1, 10^4\}$	24
N2(b)	C1 base at 5 dates	Gurobi barrier, Clarabel, SCS; risk-row normalization and objective scaling on/off; reference	65
N3	the N1 instances at default presolve	root only, both perspective arms; MIQCPMethod -1, 0, 1 (default scaling also FeasibilityTol 10^{-9})	252

$\times$ three alphas $\times$ 12 dates: (i) all six factor exposures zero, (ii) the same plus dollar neutrality, each with the risk budget and no other rows or costs. N counts names (the top N by ADV); $K = \mathrm{round}(fS)$, half up, with S the support size of P-RB at $u = 6\%$ on that universe.

B. Algorithms

P-RB is solved as the mean-variance QP $\max \alpha^\top w - \lambda w^\top \Sigma w -$costs by Gurobi's dual simplex (tolerances 10^{-9}; exact active sets and duals), with λ found by Brent's root-finding method on $\log \lambda$ so that ex-ante volatility meets the budget to relative 10^{-10}; A1 makes this exact. Clarabel's primal–dual interior-point method [17] (tolerances 10^{-12}) solves the cone form of (1) as a cross-check, and the perspective relaxation of (2) ($z \in [0, 1]$), whose finite differences give its slope. Dual simplex solves the big-M relaxation and the fixed-integer QP (P-MI with z fixed at the incumbent). The *plain relaxations* are the $z \in [0, 1]$ relaxations solved as continuous problems outside the MIP (big-M by Gurobi's dual simplex, the perspective by Clarabel), so without MIP presolve or root cuts. Gurobi's branch and bound [18] solves P-MI until the relative gap between bound and incumbent is below *MIPGap* or its *Work*, a deterministic effort count in Gurobi's work units, reaches $W^* = 340$ ($2W^*$ in I1 and I2): the Work of 300 s at the design pilot's median rate of 1.12 units/s, to two significant figures. Its MIQCPMethod parameter (how cones are handled at the nodes) was left at the default, which N3 identifies as outer approximation at most instances. The P-MI objective is multiplied by $s_{\mathrm{obj}} = 10^4$ to be of order one against the tolerances; values are reported in original units. The incumbent is *polished*: the fixed-integer QP at its z is re-solved at tolerances 10^{-9}, giving V. N2(b) runs Gurobi's barrier and CVXPY's [19] installed conic solvers, Clarabel and SCS (ECOS is not installed), at default tolerances against Clarabel at 10^{-10}.

C. Analysis

A1 (risk budget and mean-variance). With multiplier λ for the risk row and ν for the mandate rows, the Lagrangian of (1) is $\alpha^\top w -$ costs $- \lambda(w^\top \Sigma w - \sigma_m^2) - \nu^\top$(mandate), whose stationarity condition is that of the mean-variance problem at the same λ. Slater's condition holds ($w = 0$ strictly satisfies the risk row; the other rows are affine), so the KKT conditions are necessary and sufficient, and the mean-variance optimum at $\lambda = \lambda^*$ solves P-RB: the risk budget is an ε-constraint with multiplier λ^*.

A2–A3 (equality case). With only the rows $Aw = 0$, no costs and no other rows, and a binding risk row, stationarity gives $w^* = \Sigma^{-1} g/(2\lambda)$ with $g = \alpha - A^\top \nu$; $Aw^* = 0$ gives $\nu = (A\Sigma^{-1}A^\top)^{-1} A\Sigma^{-1}\alpha$; the binding row gives $\lambda = \sqrt{g^\top \Sigma^{-1} g}/(2\sigma_m)$ and $A^* = \sigma_m \sqrt{g^\top \Sigma^{-1} g}$. Divided by the unconstrained $\sigma_m \sqrt{\alpha^\top \Sigma^{-1} \alpha}$, the retained fraction $\sqrt{g^\top \Sigma^{-1} g / \alpha^\top \Sigma^{-1} \alpha}$ equals TC at w^*, and it is at most one by Cauchy–Schwarz in the Σ^{-1} metric, since g is the Σ^{-1}-orthogonal projection of α onto $\{v : A\Sigma^{-1} v = 0\}$ along $\mathrm{range}(A^\top)$.

A4 (alignment). By Woodbury, $B^\top \Sigma^{-1} = F^{-1}(F^{-1} + B^\top D^{-1} B)^{-1} B^\top D^{-1}$, so $B^\top D^{-1} \alpha_\perp = 0$ (the regression residual) gives $B^\top \Sigma^{-1} \alpha_\perp = 0$ and $\Sigma^{-1}\alpha_\perp = D^{-1}\alpha_\perp$: under the risk budget alone $\alpha_\perp$'s optimum has zero factor exposure, and factor bounds have zero shadow price. With $A = B^\top$, $g = \alpha_\perp$: the projection of A2 is unique, and $\alpha_\perp$ satisfies both $B^\top \Sigma^{-1} \alpha_\perp = 0$ and $\alpha - \alpha_\perp = \alpha_\parallel = B\gamma \in \mathrm{range}(B)$ (γ the regression coefficients). The cross term $\alpha_\perp^\top \Sigma^{-1} \alpha_\parallel$ vanishes, so $\alpha^\top \Sigma^{-1} \alpha = \alpha_\perp^\top D^{-1} \alpha_\perp + \alpha_\parallel^\top \Sigma^{-1} \alpha_\parallel$, and full factor neutrality retains $\sqrt{1 - \varphi}$ of the alpha.

A5 (envelope). By Slater's condition (A1) strong duality holds. A^* as a function of the right-hand sides is the perturbation function of a convex program, hence concave, and nondecreasing in each bound because loosening it enlarges the feasible set [12]. Wherever the multipliers are unique (for example under LICQ), $dA^*/dr_k = \mu_k$, also across nondegenerate active-set changes, where μ_k has a kink but A^* does not; by the chain rule a parameter p that moves several right-hand sides has $dA^*/dp = \sum_k \mu_k \, \partial r_k/\partial p$, the family dual of Sec. II-D.

A6 (perspective tightness). For $z \in (0, 1]$, $s \geq w^2/z \geq w^2$, so in the relaxation the perspective's risk term is at least big-M's and its value at most big-M's. The perspective of a convex function is convex, and the perspective reformulation is the closure of the perspective cuts [7], [8].

A7 (one-asset toy). For $\max aw - \lambda d w^2$ over $w \in \{0\} \cup [\ell, u]$, $w \leq b$, the value is 0 for $b < \ell$ and jumps by $a\ell - \lambda d \ell^2$ at $b = \ell$, which is positive only if $a\ell > \lambda d \ell^2$. Below ℓ the true local slope is 0, and so is the fixed-integer dual; the big-M

relaxation's slope is $a - 2\lambda d b$, and the perspective relaxation's (minimizing s over $z \in [w/u, w/\ell]$) is $a - \lambda d \ell$. For $\ell/2 < b < \ell$, $a - 2\lambda d b < a - \lambda d \ell$, so where both slopes are positive big-M's is the closer to 0, although the perspective bound is the tighter. A tighter bound need not give a better slope.

A8 (integer value functions). The P-MI value $U^*(b)$ is the upper envelope, over support patterns, of concave pieces: nondecreasing by feasible-set inclusion, in general neither concave nor continuous. The relaxation dual is the slope of an upper-bounding concave function, the fixed-integer dual the slope of the active piece, and a re-solve gives a secant across pieces. With V the polished value, $\bar{U}$ Gurobi's final bound and $\bar{U}_{\text{eff}} = \max(\bar{U}, V) + 10^{-6}|V|$ (the slack covers Gurobi's tolerances), $V \le U^* \le \bar{U}_{\text{eff}}$ at each of $b \pm \delta$ (δ absolute here; the experiments state it as a percentage of b), which gives the certified interval

$$L = \frac{V_+ - \bar{U}_{\text{eff},-}}{2\delta} \le \frac{U^*_+ - U^*_-}{2\delta} \le \frac{\bar{U}_{\text{eff},+} - V_-}{2\delta} = H, \quad (3)$$

with point slope $s = (V_+ - V_-)/(2\delta)$. A case is *certified binding* iff $L > 0$ (as distinct from *binds*, $e_p \ge 10^{-4}$, in the convex problems); $H < 0$ is impossible at true optima and flags solver error. In I1 and I2 both formulations are solved at every point, V is the larger polished value and $\bar{U}$ the smaller bound.

A9 (scaling). An absolute tolerance τ on a quantity of size m is a relative error τ/m, so rescaling a row or the objective gives an equivalent problem with different effective tolerances. In the perspective cones $w_i^2 \le s_i \zeta_i$ a held name has $\ell^2 \le s_i \le u_i^2$, so Gurobi's absolute feasibility tolerance 10^{-6} is a relative error of up to $10^{-6}/\ell^2 = 0.16$ on the smallest cones; written in each name's bound it is at most $10^{-6}(u_i/\ell)^2 = 5.8 \times 10^{-4}$. This accounts for the incumbents' cone violation, whose objective effect $\lambda^* \sum_i d_i \max(0, w_i^2/\zeta_i - s_i)$ is reported relative to $|V|$, and for certified gaps above MIPGap. It does not account for the default-scaling root bound, which N3 traces to Gurobi's outer approximation of the cones stopping before it converges. The two mechanisms are kept apart.

D. Finite differences and noise

The perspective relaxation's slope at b is the central difference $\hat{s} = (R(b + \delta_r) - R(b - \delta_r))/(2\delta_r)$ of its Clarabel values R, $\delta_r = 10^{-3}\, b$; the value is the objective at Clarabel's (w, ζ) with each s_i set to w_i^2/ζ_i, so every cone holds exactly. A relaxation fails if its residuals exceed 10^{-8} or its duality gap exceeds $\varepsilon = 10^{-9} \max(|R|, |R_{\text{ref}}|)$, R_{ref} the value at the unperturbed bound, which bounds a difference's noise by $\eta = \varepsilon/\delta_r$. The one-sided slopes $s_+ = (-3R_0 + 4R_1 - R_2)/(2\delta_r)$ and $s_- = (3R_0 - 4R_{-1} + R_{-2})/(2\delta_r)$, $R_j = R(b + j\delta_r)$, are recorded too; $s_+ - s_-$ has absolute stencil coefficients summing to 16 over $2\delta_r$, so noise alone moves it by at most 8η, and a case is flagged as a kink if $|s_+ - s_-| > \max(10^{-4}|\hat{s}|, 8\eta)$. The *gap* of a relaxation at b is $(R - V)/|V|$.

E. Validation

521 passing tests and logged checks, each with a negative control, confirm the analysis: the root-found QP and the Clarabel QCQP agree on A^* to 1.2×10^{-10} and on λ^* to 3.8×10^{-8} (A1); A2–A3 hold to 8.9×10^{-16} and the A4 split to 2.1×10^{-16}; on 202 one-asset toys (two fixed and 200 seeded random ones; 8279 grid points) both formulations match A7 to 1.4×10^{-17}, the 3 points within Gurobi's tolerance below ℓ excluded and counted; certified intervals contain the exact slope on 4 toy and 4 small MIQP cases; P-MI with $K \ge N$, $\ell = 0$ equals P-RB to 1.5×10^{-15}. An independent feasibility checker passes every C1 and C2 solution and every polished V. The slope method was gated before I1 on the 200 random toys (0 misses at 8399 points) and 39 objective-scale pairs (0 outside tolerance).

IV. Results

A. Q1: Constraint Costs in the Convex Problem

Setup. C1 and C2 (Table II) at the D12 dates on all eligible names. The criterion is the elasticity e_p from exact duals, validated as derivatives below. Statistics are over the D12 dates for α_{mom} ($n = 12$) unless stated.

Which constraints bind. At base, A^* has median 30.5 bp/month (range 22.0–38.3), a median 0.65 (0.51–0.77) of the unconstrained, costless optimum. The risk budget binds in all but 4 of the 36 base solves, all of them $\alpha_{\|}$. The liquidity cap binds at all 12 dates (median elasticity 0.16), sector bounds at 12 (median 0.014), dollar neutrality at 10 (1.3×10^{-3}), style bounds at 4 and the position cap at 2; beta and gross bind at none. Fig. 1 shows the four families that cost the most. At the tightest grid points the pod keeps a median 0.744 of its alpha at $\theta = 2.5\%$ (range 0.52–0.95), 0.961 under exact sector neutrality and 0.956 with every style exposure at zero. The other families are flat: exact dollar neutrality keeps 0.997, exact beta neutrality 0.999, each other style factor alone at zero a median of at least 0.997, gross $G = 1$ 0.994 and $u = 1.5\%$ 0.997.

What determines the cost. Factor bounds are cheap because α_{mom} is weakly aligned with the factors: φ is 0.011–0.070 over D12, and under idealized full factor neutrality the retained fraction equals $\sqrt{1 - \varphi}$ (A4), at least 0.964 (Fig. 2). Numerical and closed-form fractions agree to 7.5×10^{-8} over the 36 mandate-(i) solves, and A^* to 1.6×10^{-14} relative over the 48 solves where the closed form is nonzero and to 3.6×10^{-17} absolute over the 24 $\alpha_{\|}$ solves where it is zero. $\alpha_{\perp}$ ($\varphi \le 3.1 \times 10^{-33}$) loses nothing and $\alpha_{\|}$ ($\varphi = 1.0$) everything. At the base mandate the α_{mom} points lie well below the curve: the cost comes from the caps and the costs. Of the base portfolio's gross expected alpha $\alpha^\top w^*$, the pod keeps a median 0.80 (0.74–0.84) as A^*, pays 0.16 (0.13–0.21) in trading and 0.039 (0.032–0.052) in borrow.

Duals as derivatives. At every binding key and D12 date (115 checks over 18 keys; each sector and style factor is a key beside the families) the dual matches the central difference at $\delta = 1\%$ of the bound within 1% (largest error 3.1×10^{-4}), rechecked at 0.1% at the 64 flagged stencils (the dual zero at an end, or forward and backward differences more than 1% apart; 35 of them change the active set, 29 are curvature flags);

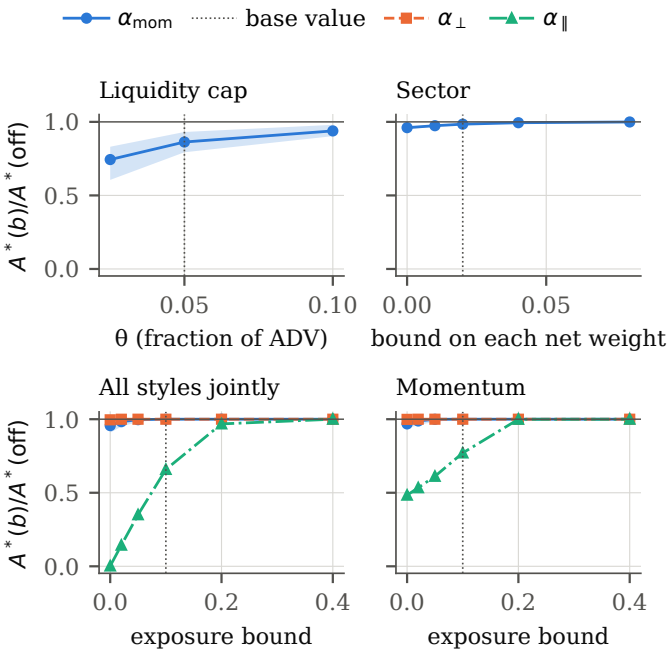


Fig. 1. Cost curves (C2): retained alpha $A^*(b)/A^*(\text{off})$, with $A^*(\text{off})$ the value with that family removed, against the bound; all other constraints at base (dotted: base value). Solid: the α_{mom} median with its interquartile band over the 12 D12 dates; on the factor panels the $\alpha_\perp$ (dashed) and $\alpha_\parallel$ (dash-dotted) medians.

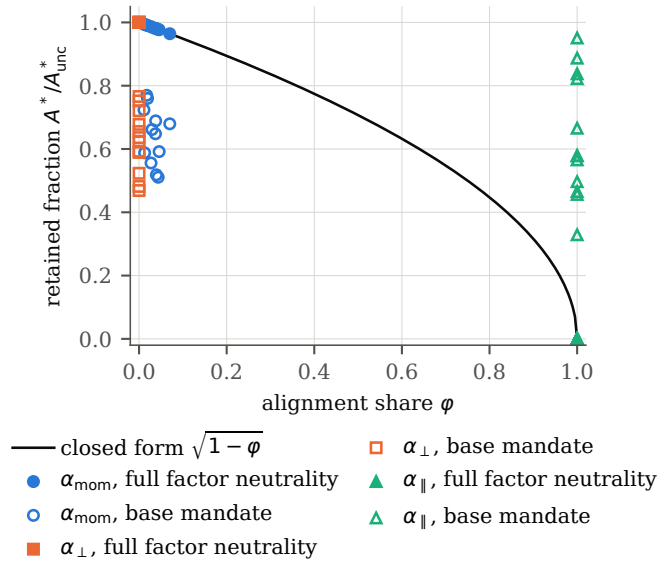


Fig. 2. Retained fraction A^*/A^*_{unc} against the alignment share φ (C1). Curve: the closed form $\sqrt{1-\varphi}$ (A4). Filled: idealized full factor neutrality, mandate (i) (three alphas $\times$ 12 dates). Hollow: the base mandate. Both relative to the unconstrained, costless P-RB.

along all 312 C2 sweeps A^* is nondecreasing and concave in the bound (0 failures), as A5 predicts.

B. Q2: Sensitivity under Integrality

Setup. I1 (Table II) perturbs four bounds by δ (sector and dollar at 2%, all styles at 0.10, and the momentum bound

TABLE III
I1 ESTIMATOR ERRORS $|\hat{\mu}-s|/|s|$ AT $\delta = 10\%$ OVER CERTIFIED-BINDING CASES, BY f, WITH THE NUMBER IN $[L, H]$ AND THE MEDIAN RELATIVE INTERVAL WIDTH. BIG-M DUAL: OF THE BIG-M RELAXATION; PERSPECTIVE: THE RELAXATION'S SLOPE. CONVEX BASELINE: ITS DUAL AGAINST ITS OWN SECANT.

estimator	f	n	median	max	in $[L, H]$	med. $(H-L)/s$
big-M dual	0.25	7	2.6×10^{-1}	4.2×10^{0}	0	6.2×10^{-4}
big-M dual	0.5	8	1.4×10^{-1}	1.2×10^{0}	0	3.2×10^{-4}
big-M dual	0.75	8	1.9×10^{-2}	3.0×10^{-1}	0	3.2×10^{-4}
perspective	0.25	7	4.5×10^{-2}	8.2×10^{-1}	0	6.2×10^{-4}
perspective	0.5	8	8.2×10^{-3}	1.0×10^{-1}	0	3.2×10^{-4}
perspective	0.75	8	8.4×10^{-3}	5.2×10^{-2}	0	3.2×10^{-4}
fixed-integer	0.25	7	4.8×10^{-2}	7.8×10^{-1}	1	6.2×10^{-4}
fixed-integer	0.5	8	6.0×10^{-2}	8.2×10^{-1}	0	3.2×10^{-4}
fixed-integer	0.75	8	5.2×10^{-3}	2.9×10^{-2}	3	3.2×10^{-4}
convex baseline	all	24	3.7×10^{-4}	4.2×10^{-3}	–	–

tightened to 0.025): 48 cases, 36 at $\delta = 10\%$. The estimators are the big-M relaxation dual, the perspective relaxation slope and the fixed-integer dual; the *continuous baseline* is P-MI with $\ell = 0$ and no cardinality row. The criterion is the relative error $|\hat{\mu}-s|/|s|$ over certified-binding cases, and whether $\hat{\mu} \in [L, H]$.

Certification. No interval is violated (0 of 144 points) and the formulations' polished values tie at all 144. At $\delta = 10\%$, 23 of 36 cases are certified binding (sector 9, momentum 6, dollar 5, all styles 3); in the others every estimator is zero (the perspective slope to within 1.4×10^{-11}). The median relative width $(H-L)/s$ is 3.6×10^{-4} (1.2×10^{-4}–0.020, $n = 23$): lying in $[L, H]$ means matching the secant to that width.

Estimator errors. Table III gives the errors at $\delta = 10\%$ ($n = 23$). The perspective slope has the smallest median error, 0.013 (max 0.82), then the fixed-integer dual, 0.023 (max 0.82), and the big-M dual, 0.073 (max 4.2). Only 4 estimates, all fixed-integer duals, lie in $[L, H]$. The baseline row counts the 24 cases where the baseline's own constraint binds ($e \geq 10^{-4}$): the 23 certified-binding cases and 1 more. Its dual is within a median 3.7×10^{-4} (max 4.2×10^{-3}) of its own secant, so integrality, not the step, causes the errors. The errors depend on δ: at 2017-12 ($n = 11$ binding at each step) the medians at $\delta = 5\%$ and 10% are 0.014 and 0.023 (fixed-integer), 0.052 and 0.016 (perspective), 0.24 and 0.073 (big-M).

When the fixed-integer dual is exact. Each case is classed by comparing, at $b-\delta$, b and $b+\delta$, the signed support of V's incumbent and the fixed-integer QP's active set: P1, both unchanged; P2, support unchanged, active set changed; P3, support changed. Of the 34 certified-binding cases (both steps pooled), 3, 2 and 18 are P1, P2 and P3 at $\delta = 10\%$. With a fixed support and active set the QP solution is affine in b, so the value is quadratic and the central secant equals the derivative at b: in P1 the fixed-integer error is at most 6.7×10^{-12} ($n = 7$; predicted $\leq 10^{-6}$), in P2 at most 2.9×10^{-4} ($n = 3$), and in P3 its median is 0.056 ($n = 24$). The relaxation slopes fail differently: their error rises with the relaxation's gap at b

(Spearman rank correlation 0.64 for the perspective, 0.50 for big-M, $n = 23$).

Value functions. I2 sweeps the sector bound, the I1 constraint with the largest median secant (1.364×10^{-3}, against 1.316×10^{-3} for the momentum bound). The P-MI value lies below the baseline value and is not concave: 12 of 39 grid triples at 2017-12 and 9 of 39 at 2020-03 are certified nonconcave, $V(b^{(k-1)}) + V(b^{(k+1)}) > 2\bar{U}_{\text{eff}}(b^{(k)})$, and the support changes between 16 of 40 and 17 of 40 neighbouring grid points $b^{(k)}$. Fig. 3 shows the slopes. At $0.5b$, b and $1.5b$ ($\delta = 10\%$, $n = 6$) the median errors are 0.32 (big-M), 0.078 (perspective) and 0.059 (fixed-integer), none in its interval (relative width at most 9.9×10^{-4}). The perspective slopes agree at $10s_{\text{obj}}$ on all 12 gated and 36 of the 36 other I1 cases, and none of the 240 I1 relaxations failed.

C. Q3: Formulations and Numerics

Setup. N1 (Table II) solves the 12 instances per tightness level (loose: styles 0.4, beta 0.2; base; tight: 0.025, 0.0125) with big-M, the perspective at default scaling and the rescaled perspective, at MIPGap 10^{-4} and WorkLimit W^*. The criteria are Work and the root gap, (root bound − best known value)/|best known value|, taken after Gurobi's root cut loop. Default-scaling results are reported as solver behaviour: they describe how Gurobi treats the unscaled cones, not the formulation.

Root gaps and effort. At default presolve (Fig. 4) the rescaled perspective has the smallest median root gap at loose and base tightness and big-M at tight (rescaled 4.1×10^{-4}, 3.3×10^{-4}, 1.3×10^{-3}; big-M 4.7×10^{-4}, 4.4×10^{-4}, 1.1×10^{-3}), and the least median Work at every level (0.66, 0.66, 0.77; big-M 1.1, 1.0, 1.1). The default-scaling perspective is far behind (0.011, 0.015, 0.022; Work 1.4, 1.2, 2.1). No solve at default presolve stopped at the WorkLimit. The plain relaxations order as A6 predicts on every instance (0 violations; median margin 0.057 of $|V|$), with median plain gaps 0.059 (big-M) and 1.8×10^{-3} (perspective) at base. Presolve, not the formulation, closes big-M's gap: at $N = 250$ ($n = 12$ per arm) turning it off raises big-M's median root gap from 6.0×10^{-4} to 0.057 (its plain gap is 0.060), and 6 of 12 solves stop at the WorkLimit, the others using 4.3–171 Work. The rescaled perspective moves only from 4.9×10^{-4} to 5.5×10^{-4} (median Work 0.766 to 2.28).

Two mechanisms. The default-scaling root bound is not the cone tolerance. N3 (Table II) repeats the root of both perspective arms at every N1 instance at default presolve; the default probe reproduces N1's root bound exactly at 36 of 36. Forcing outer approximation (MIQCPMethod 1; the default, −1, lets Gurobi choose) gives the same bound exactly at 29 of 36 (rescaled: 30), so Gurobi chose outer approximation at most instances. A QCP node relaxation (method 0) lowers the median default-scaling root gap from 0.015 to 5.7×10^{-4}, near the plain perspective relaxation's 1.9×10^{-3} (within $5.0 \times 10^{-3}|V|$ at 33 of 36), while FeasibilityTol 10^{-9} leaves it at 0.015 (median ratio 0.98; 2 instances collapse). The rescaled arm's gaps under the three methods stay within a factor 7.7 of each other. N3 was added after the other experiments had been analysed, to test on all 36 instances what the design pilot (S2d) had found on eight of them. The incumbent mechanism is the cone tolerance (A9): at default presolve ($n = 36$ per arm) the default-scaling incumbents violate their cones within tolerance with an objective effect of median 4.6×10^{-4} and max 6.4×10^{-3} of $|V|$, at most 1.9×10^{-11} rescaled. The rescaled arm changes both the rows' scale (divided by $\max_i u_i$) and the cones' scale. Every arm's polished value lies within every other arm's $\bar{U}_{\text{eff}}$ (0 failures in 144 pairs).

TABLE IV
N2(B): P-RB AT THE DATES 2008-10, 2013-06, 2017-12, 2020-03, 2025-04, SOLVERS AT DEFAULT TOLERANCES AGAINST CLARABEL AT 10^{-10}, WORST OVER THE DATES AND BOTH RISK-ROW NORMALIZATIONS; n SOLVES PER ROW. DUAL ERR.: WORST RELATIVE ERROR IN A BINDING DUAL.

solver, objective	n	$\lvert\Delta A^*\rvert/A^*$	$\lVert\Delta w\rVert_1/\lVert w\rVert_1$	dual err.
Gurobi barrier unscaled	10	3.1×10^{-5}	1.9×10^{-3}	5.7×10^{-2}
Gurobi barrier scaled	10	7.2×10^{-7}	1.2×10^{-3}	9.5×10^{-3}
Clarabel unscaled	10	2.7×10^{-7}	7.4×10^{-5}	2.5×10^{-4}
Clarabel scaled	10	1.7×10^{-8}	1.3×10^{-5}	2.2×10^{-4}
SCS unscaled	10	5.1×10^{-4}	5.6×10^{-4}	2.7×10^{-3}
SCS scaled	10	2.7×10^{-6}	2.9×10^{-5}	1.0×10^{-4}

Objective scaling. N2(a) repeats N1's base instances at $N = 250$ at $s_{\text{obj}} = 1$ and 10^4. At $s_{\text{obj}} = 1$ big-M stops at the WorkLimit in 2 of 4 instances (the others use 2.9–4.7 Work), and the rescaled perspective at 2017-12 with $K = 67$ reported a closed gap while its $\bar{U}_{\text{eff}}$ lay $9.1 \times 10^{-5}|V|$ below big-M's polished value, a false certificate. At 10^4 no incumbent exceeds another arm's bound (0, against 1), and the default-scaling arm's median certified gap is 1.2×10^{-3}, above MIPGap 10^{-4}: its cone effect reaches 4.8×10^{-3} of $|V|$, while the rescaled arm's median gap is 9.0×10^{-5}. In N2(b) (Table IV) objective scaling cuts the barrier's errors from 3.1×10^{-5} to 7.2×10^{-7} in A^* and 0.057 to 9.5×10^{-3} in the duals; risk-row normalization leaves its A^* identical (10 of 10).

V. DISCUSSION

TODO(Isaiah), V, 0.8 page; the coverage list of modeling choices, limitations, trade-offs, unexpected behavior and extensions (claim → evidence):

- Modeling choice, fixed risk budget against fixed λ: costs at equal risk; mean-variance recovered at λ^* (A1) → 3.8×10^{-8}.
- Modeling choice, λ^* fixed within (date, N) for P-MI: exact only without integers; the MIQP's volatility falls below the budget (I1 median 2.87%, all 48 of 48 base points below; N1 4 of 36 above), so the Q2 slopes are of the penalized objective → Sec. II, the baseline at 39 of 48 I1 base points on the budget.
- Modeling choices: single period with $w_0 = 0$; the six-factor, 252-day risk model with the residual floor; the momentum construction and winsorization; the base mandate values; the 6% cap and ℓ in P-MI; costs as assumptions → Table I.
- Limitations: the disclosures below.

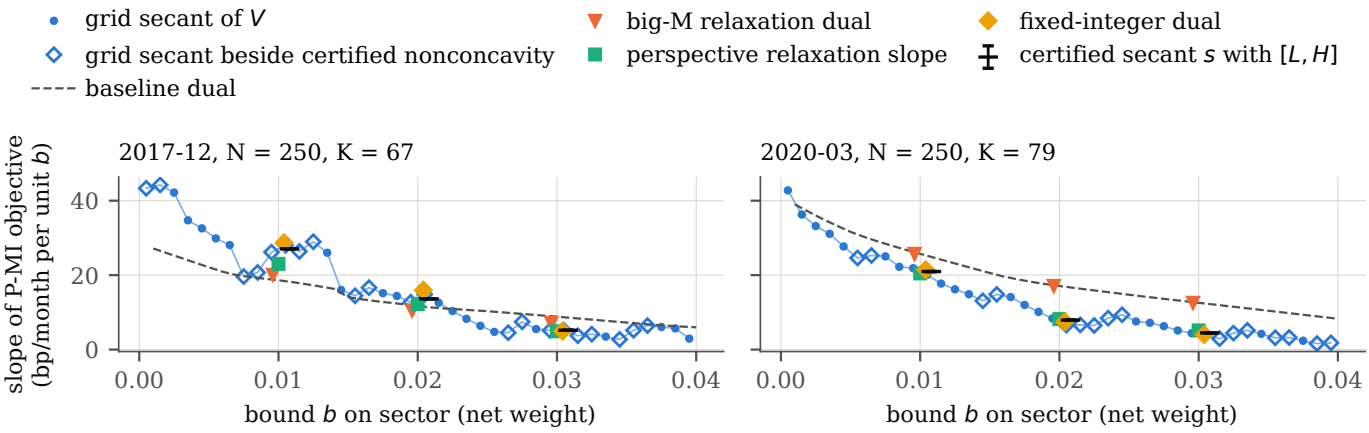


Fig. 3. I2 slopes on the sector bound, 250 names (left 2017-12, right 2020-03), as P-MI objective (alpha minus λ^* times variance minus costs) per unit of bound: the secant of V between neighbouring grid points (hollow diamonds beside a certified nonconcave grid point), the baseline's dual, and at $0.5b$, b and $1.5b$ the three estimators beside the certified secant s with $[L, H]$ (too narrow to see; offset slightly for legibility).

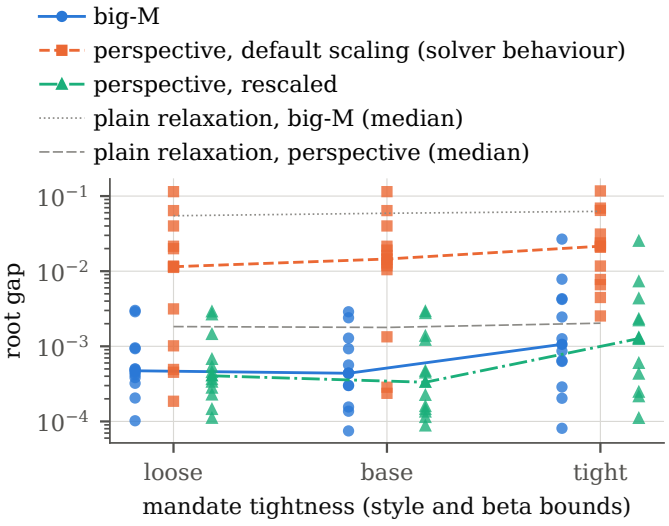


Fig. 4. N1 root gap after the root cut loop against mandate tightness, default presolve; one marker per instance ($n = 12$ per cell), medians connected. The default-scaling perspective is solver behaviour. Thin gray lines: the median plain-relaxation gaps.

- Trade-off, alpha against exposure: the cost curves → Fig. 1, 0.744, 0.961.
- Trade-off, formulation strength against per-node cost: tighter at the root, but cone nodes → 0.057; median nodes at base 36.5 (big-M), 6.5 (rescaled), 270.5 (default).
- Trade-off, tightness against effort → Fig. 4, 1.1, 0.77.
- Trade-off, tolerance against runtime: MIPGap 10^{-6} (I1) against 10^{-4} (N1); at $s_{\text{obj}} = 1$ big-M stops at the WorkLimit in 2 of 4 (others 2.9–4.7 Work) → Sec. IV-C.
- What each estimator measures (A8), and why the errors depend on δ → 0.052 against 0.016 at 2017-12; at $\delta = 5\%$ the P1 maxima are 3.6 (perspective) and 5.4 (big-M), $n = 4$.
- Two failure modes: a support change within the step for the fixed-integer dual (P3 0.056; P1 6.7×10^{-12}), and relaxation looseness for the relaxation slopes (Spearman 0.64, 0.50; perspective per class 0.011, 0.014) → Sec. IV-B.
- The I2 near-tie: median secant 1.364×10^{-3} (sector) against 1.316×10^{-3} (momentum); under the perspective slope the order flips, 1.222×10^{-3} against 1.306×10^{-3} → Sec. IV-B.
- Unexpected: presolve does big-M's root work → 6.0×10^{-4} against 0.057; 6 of 12 stopped at the WorkLimit with presolve off.
- Unexpected: the default-scaling root bound is outer approximation stopping early, not the cone tolerance → 29 of 36, 5.7×10^{-4}, 0.98.
- Unexpected: the false certificate at $s_{\text{obj}} = 1$ → 9.1×10^{-5}.
- Unexpected: the risk-row normalization is absorbed by presolve (supported, not proven) → 10 of 10, 1.6×10^{-5}.
- Unexpected: the hedged legs (in 7 of 45 perspective relaxations the relaxation splits a name into a long and a short leg to cheapen its cone; 11 name–relaxation pairs, 4 distinct names, the prediction holding at 11); per-leg cones as an extension → 11, 4, 11 (not yet in Results).
- Slope accuracy holds on every I1 case, not only the gated ones → 36 of 36.
- The earlier iteration's lessons in one or two sentences, without numbers: the fixed-risk formulation and the scaling policy.
- Extensions: platform coordination by exposure prices (decomposition); multi-period rebalancing (MPC); robust alpha (SOCP); a Lagrangian bound for the cardinality row.

VI. Conclusion

TODO(Isaiah), VI, 0.2 page (claim → evidence):

- Q1: neutrality is cheap for a weakly aligned alpha; the liquidity cap, sector bounds and trading costs carry the cost → 0.16, 0.16, 0.964.
- Q2: the fixed-integer dual is exact on a stable support and the best estimator depends on the step → 6.7×10^{-12}, 0.056, Table III.
- Q3: two mechanisms, outer approximation at the root and the cone tolerance at the incumbent; the perspective formulation helps only with its weights rescaled, and objective scaling is still needed → 3.3×10^{-4}, 0.015, 29, 9.1×10^{-5}.
- Extensions → Sec. V.

Disclosures and AI-Use Acknowledgment

TODO(Isaiah), disclosures, the eight required items (claim → evidence):

- Universe: current S&P 500 constituents (snapshot 2026-10-16), so survivorship and look-ahead in the universe; GICS sectors are not point-in-time → results/data_report.md.
- Yahoo adjusted data; the JCI and DHR errors masked; a day is valid only with a return and dollar volume of at least $1M, a stock eligible with 80% valid days in its window; 1 multi-session return enters the regressions as a one-day return → results/data_report.md.
- One-pass winsorization leaves $|z|$ above 3: max 5.47 (2024-04), at 240 of 240 month-ends, 1936 of 107592 name-dates → results/disclosures/.
- Single period, $w_0 = 0$; borrow and trading costs are assumptions → Table I.
- No backtest and no performance claims; the alpha is a simple input signal.
- The parameter-setting rules of the design pilot (s_{obj}, W^*, the K grids, the I1 constraints) → Sec. III, experiments/configs/integer.yaml.
- The earlier iteration, without numbers.

TODO(Isaiah), AI-use acknowledgment, from docs/AI_USE.md's v2 log (claim → evidence):

- Claude (chat) designed v2 and extracted the repository; proposed most design decisions, check text, design edits and gate notes; sketched derivations; drafted the proposal text. Isaiah chose among the proposals, edited and signed every gate note, and ran the commands.
- Claude Code, with the result-checker and python-reviewer subagents, wrote and ran the code, tests and checks, built the figures and tables, and drafted Sections II–IV of this report (Mode A); Isaiah revises them and checks the derivations by hand.

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